A2: Second Order Type-II Integer-N Phase Locked Loop Analysis, Design, and Simulation

EN4431: Analog IC Design

March 30, 2026

CMOS phase-locked loops (PLLs) are widely used in most of the system-on-chips (SoCs) as the clock generator for digital circuits and wireline transceivers, or the frequency synthesizer for the wireless transceivers. Clock frequencies and data rates have been increasing with each generation of processing technology and processor architecture. Within these systems, well-timed clocks are generated with PLLs and then distributed on-chip with clock buffers. The rapid increase of the systems' clock frequency poses challenges in generating and distributing the clock with low uncertainty and low power. Analog and mixed-signal PLL (AMS-PLL) is the most widely used architecture for clock generators and frequency synthesizers.

Figure 1(a) shows the block diagram of the basic PFD-CP (Type-II) PLL, which is an implementation of an AMS-PLL architecture. It consists of a tri-state phase/frequency detector (PFD), a charge pump

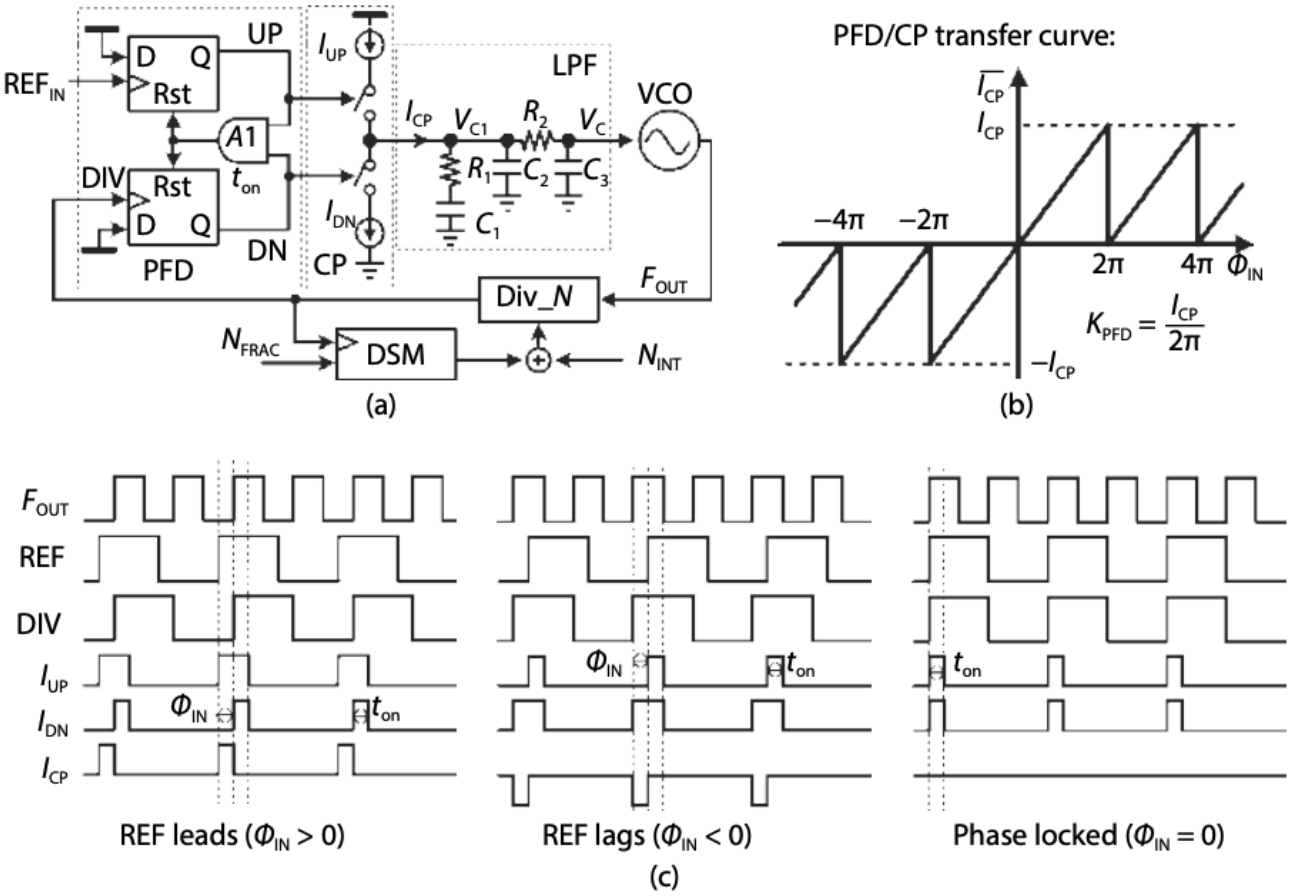


Figure 1: (a) A basic PFD-CP (Type-II) PLL (b) Transfer curve of PFD and CP. (c) Timing diagram.

(CP), a low-pass filter (LPF), a voltage-controlled oscillator (VCO) and a divide-by-N frequency divider (Div_N). The ratio between the VCO output frequency (f_{VCO}) and the reference frequency (f_{REF}) is N; i.e. $N = \frac{f_{VCO}}{f_{REF}}$. The delta–sigma ($\Delta\Sigma$) modulator (DSM) is used to modulate the division ratio of the divider so as to obtain a fractional division ratio to achieve the function of the fractional-N PLL [1]. For the integer-N PLL, of which N is a rational number, the DSM is not required.

The operation principle of the PFD-CPPLL is presented by the timing diagram shown in Figure 1(c). As an example, the division ratio N shown in Figure 1(c) is 2. Φ_{IN} shown in Figure 1(c) is the input phase error between the reference clock (REF_{IN}) and the divider feedback clock (DIV).

- When the PLL is unlocked and Φ_{IN} is positive, Φ_{IN} is detected by the PFD and CP generates a positive current pulse I_{CP} to charge the LPF so as to increase f_{VCO} to reduce Φ_{IN}.
- When the PLL is unlocked and Φ_{IN} is negative, CP generates a negative current pulse I_{CP} to discharge the LPF so as to decrease f_{VCO} to reduce $|\Phi_{IN}|$.
- At the locking state, Φ_{IN} keeps zero so that the VCO tuning voltage V_C keeps stable. As a result, $f_{VCO} = N \cdot f_{REF}$ (N shown in Figure 1(c) is 2).

Unlike the PLLs with other types of the phase detector (PD) such the XOR gate and mixer, which suffer from the issue of limited frequency lock range due to the limited monotonic Φ_{IN} range of these PDs, the lock range of the PFD-based PLL is unlimited because the linear Φ_{IN} range of the tri-state PFD is from -2π to 2π (see Figure 1(b)), and the PFD output polarity keeps positive or negative when $\Phi_{IN} > 2\pi$ ($f_{REF} > \frac{f_{VCO}}{N}$) or $\Phi_{IN} < -2\pi$ ($f_{REF} < \frac{f_{VCO}}{N}$), respectively. This makes the PFD be able to distinguish the polarity of the frequency difference between f_{REF} and ($\frac{f_{VCO}}{N}$) so that an additional frequency locked loop with a separated frequency detector for initial frequency acquisition is not required. Hence, the PFD-CPPLL architecture is simple and robust, and thus it is the most widely used AMS-PLL architecture [2]. Additional information regarding PLLs is available in [3].

1 Theoretical Analysis

Before designing in the SKY130 process, you must mathematically analyze the PLL as an abstract model and find out the requirements of each block of the system. You will be implementing the Second Order Type-II Integer-N PLL in Figure 2 for a Tapeout as a part of this EN4431 - Analog IC Design course.

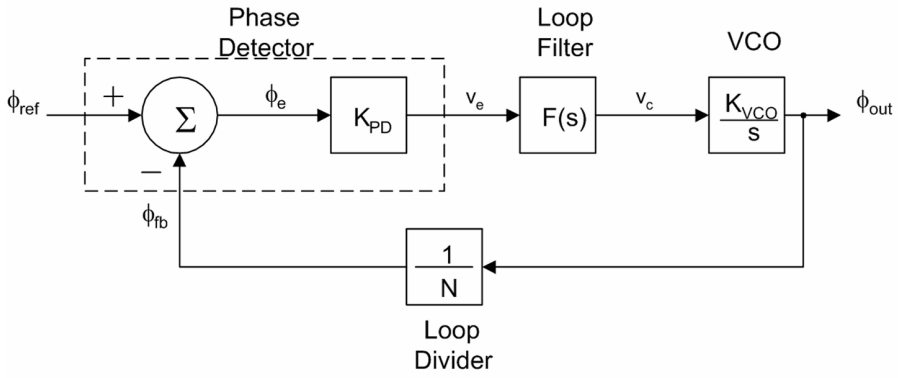


Figure 2: Abstract block diagram of a Second-order Type-II Integer-N PLL [4].

1.1 Transfer Function Derivations and Analysis

In a second-order PLL, stability depends on both the loop filter $F(s)$ and the system constants (K_{PD}, K_{VCO}, and N). While the loop filter shapes the phase profile to ensure stability, the constants determine

the overall loop gain. This gain sets the crossover frequency, which dictates the actual phase margin the system operates with. If the gain is too high or too low, the crossover frequency shifts to a point with poor phase margin, leading to instability.

Q1.1.A In most cases, the transfer function of both the PFD and CP of a PLL system is expressed as K_{PD}. We can approximate the PFD/CP circuit of Figure 3 as a current source of some average value driving C. For an input phase difference of $\Delta\phi$, calculate (i) the average value of the current source (I_{avg}) and (ii) the output slope, for an input period of T_{in}.

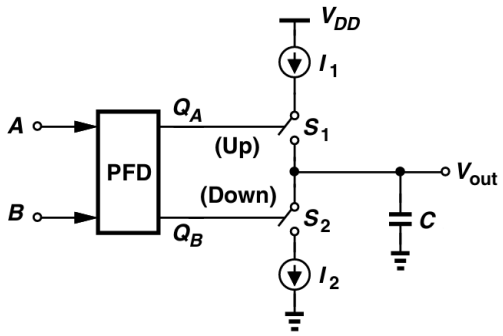


Figure 3: Cascaded PFD and CP blocks. Nominally, we design to be $I_1 = I_2 = I_{CP}$ [5].

$$\text{Finally,} \qquad K_{PD} = \frac{I_{avg} \text{ (for an input period } T_{in})}{\Delta\phi}$$

Q1.1.B To reduce the ripple of V_c, now we will replace C of Figure 3 with the loop filter design in Figure 4. Derive the transfer function $F(s)$ of this loop filter design. *Note: As the input to the block is*

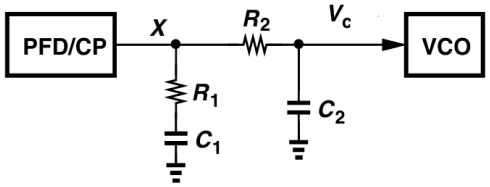


Figure 4: Second-order loop filter design [5].

current I_{CP} from the CP, and the output is voltage V_c to the VCO, $F(s)$ actually represents the impedence of the loop filter.

Q1.1.C Let $\tau_z = R_1C_1$, $\tau_p = R_2C_2$, and $C_{tot} = C_1 + C_2$. Rearrange your derivation of **Q1.1.B** and show that,

$$F(s) = \frac{1}{sC_{tot}} \cdot \frac{(1 + s\tau_z)}{(1 + s\tau_p)} \tag{1}$$

Q1.1.D Derive the open-loop transfer function, $G(s)$, by breaking the loop at the loop divider output of the PLL system in Figure 2. Write in terms of K_{PD}, $F(s)$, K_{VCO}, and N.

$$G(s) = \frac{\phi_{fb}(s)}{\phi_e(s)} \tag{2}$$

Q1.1.E Rewrite $G(s)$ by substituting K_{PD} and $F(s)$ using the answers of **Q1.1.A** and **Q1.1.C** and determine all the poles and zeros of this open-loop system.

Q1.1.D Determine the loop gain, $|G(s)|$, of the system.

Q1.1.E Loop bandwidth is defined as the crossover frequency (ω_c) at which the magnitude of the loop gain equals 0 dB ($|G(s)| = 1$). A wider bandwidth reduces locking time but may increase instability, while a narrow bandwidth improves stability. Phase margin (ϕ_m) is defined as the amount of phase you have above -180° exactly at ω_c. Hence, show that,

$$\phi_m = \arctan(\tau_z \cdot \omega_c) - \arctan(\tau_p \cdot \omega_c) \tag{3}$$

Q1.1.F [Optional] Determine the ω_c at which ϕ_m is maximum by setting, $\frac{d\phi_m}{d\omega_c} = 0$.

Q1.1.G Determine the required C_{tot} at crossover frequency, ω_c, using the loop gain equation of **Q1.1.D**.

Q1.1.H Derive the closed-loop transfer function, $H(s)$, of this PLL system in terms $G(s)$ and N.

$$H(s) = \frac{\phi_{out}(s)}{\phi_{ref}(s)} \tag{4}$$

1.2 Requirements Analysis

Recall the K_{VCO} value you obtained in your previous assignment (A1). The following required design specifications, summarized in the Table 1, are provided for this project. A generally accepted rule of

Table 1: Desired Project Performance Specifications

Parameter	**Project Specification**
Technology [Min. length of transistors (L_{min})]	150 nm (SKY130A)
Supply voltage V_{DD}	1.8 V
Operating VCO control voltage $V_{DD}/2$	0.9 V
Reference frequency (f_{REF})	2 MHz
VCO sensitivity (K_{VCO})	From A1 (MHz/V)
Charge pump current (I_{CP})	100 μA
Division ratio (N)	$25 \leq$ N ≤ 75
PLL locking frequency range (f_{OUT})	50 MHz to 150 MHz
Loop bandwidth (f_c)	$\approx$ 180 kHz
Phase margin (ϕ_m)	$\geq 60^\circ$
Expected PLL lock time (t_{lock})	$t_{lock,min}$ ps - $t_{lock,max}$ ps
Overall average DC power consumption ($P_{total,avg}$)	? μW

thumb is that the loop bandwidth (f_c) should be much less than 1/10 of the reference frequency (f_{ref}). As $f_{ref} = 2$ MHz, f_c is chosen to be 180 kHz. Also, the charge pump current I_{CP} is fixed at 100 μA.

Q1.2.A By using the equations derived in **Q1.1**, determine R_1, R_2, C_1, and C_2 so that loop bandwidth (ω_c) and phase margin (ϕ_m) requirements are achieved at the output frequency (f_{OUT}) of 100 MHz.

Q1.2.B Plot the open-loop magnitude and phase response and measure ω_c and ϕ_m to verify.

Q1.2.C After substituting all calculated values to $H(s)$, plot the closed-loop step response.

2 Design and Simulation (Using Cadence tools & SKY130 PDK)

It is recommended to refer to the following articles when realizing the required PLL blocks.

- A Ring-Oscillator-Based 2.4 GHz Integer-N PLL Design in SkyWater 130nm Open-Source Technology - ISPACS 2025 [6].
- Practical Phase-Locked Loop Design - ISSCC 2004 Tutorial [7].

2.1 Blockwise Design and Simulation

Implement each block individually and use the relevant testbenches to verify the performance. The recommended order of design is as follows.

1. Phase frequency detector (PFD)
2. Charge pump (CP)
3. Loop filter (LF)
4. Testing cascaded PFD/CP/LF with slightly different frequencies as input.
5. Programmable divider (Div_N).

2.2 Integrated Simulation and Characterization

Integrate all the blocks as shown in Figure 2 to build the complete PLL system. Use your VCO design from A1.

Q2.2.A Run a basic transient simulation to verify startup and steady-state oscillation at the target center frequency (with N set to 50). Plot the output waveform.

Q2.2.B Characterize and report the measured performance metrics of your PLL design. *Note: Try to measure all the performance metrics as reported by Assaify et al. [6].*

Instructions

Please submit your report (at least 5 pages without appendix), including the following information;

- Design methodology and details of the calculation and device sizes
- Transient simulation results at maximum output frequency (up to after lock time t_{lock}) and its frequency spectrum (FFT) showing the frequency of oscillation (f_0) and sidebands
- [Optional] Parametric simulation results of output frequency of oscillation f_{OUT} (and PLL lock time t_{lock}) vs. fractional division ratio (N). You may need to use MATLAB to obtain the exact plot.

- Discuss an improvement that can help achieve an Integer-N PLL using the architecture in Figure 1(a). (Assume that the DSM is not used.)
- Attach the circuit netlist in the appendix

For more guidance, please refer to the reference materials available in A2 drive folder. Badiger, N.A., Iyer, S. Design & Implementation of High Speed and Low Power PLL Using GPDK 45 nm Technology. J. Inst. Eng. India Ser. B 105, 239–249 (2024) [8] and also read the Design and Analysis of Efficient Phase Locked Loop (2015), Naik, M. D. Kumar [9].

Due Date: 19th April 2026 by 11.59 PM

Note - It is not essential for you to achieve the most optimal design with the best performance. If you can attempt to cover the most important simulations and relevant test benches for those simulations and present your work, it will be adequate to get the full assignment grades. Also, you have the freedom to improve the circuit as you wish with proper justifications.

References

[1] B. Razavi, "The delta-sigma modulator [a circuit for all seasons]," *IEEE Solid-State Circuits Magazine*, vol. 8, no. 2, pp. 10–15, 2016.

[2] Z. Zhang, "Cmos analog and mixed-signal phase-locked loops: An overview," *Journal of Semiconductors*, vol. 41, no. 11, p. 111402, nov 2020. [Online]. Available: https://dx.doi.org/10.1088/1674-4926/41/11/111402

[3] T. V. H. Nguyen and C.-K. Pham, "An overview of phase-locked loop: From fundamentals to the frontier," *Sensors*, vol. 25, no. 18, 2025. [Online]. Available: https://www.mdpi.com/1424-8220/25/18/5623

[4] S. Palermo, "Lecture 28: PLL Analysis," ECEN620, Texas A&M University, 2014. [Online]. Available: https://people.engr.tamu.edu/spalermo/ecen620.html

[5] B. Razavi, *RF Microelectronics*, ser. Always learning. Pearson Education International, 2012. [Online]. Available: https://books.google.lk/books?id=ovGEZwEACAAJ

[6] N. F. Assaify, A. N. Irfansyah, and T. Mujiono, "A ring-oscillator-based 2.4 ghz integer-n pll design in skywater 130nm open-source technology," in *2025 International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, 2025, pp. 1–5.

[7] D. Fischette, "Practical phase-locked loop design," Tutorial presented at the IEEE International Solid-State Circuits Conference (ISSCC), February 2004. [Online]. Available: https://www.eecis.udel.edu/~vsaxena/courses/ece518/Handouts/PLLTutorialISSCC2004.pdf

[8] N. Badiger and S. Iyer, "Design & implementation of high speed and low power pll using gpdk 45 nm technology," *Journal of The Institution of Engineers (India) Series B*, 01 2024.

[9] M. D. K. Naik, "Design and analysis of efficient phase locked loop," B.Tech. thesis, National Institute of Technology Rourkela, 2015. [Online]. Available: https://files01.core.ac.uk/download/pdf/80148289.pdf